Functional Standard Work

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Title: RLPM Production Phase

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1 Purpose

1.1 The purpose of this standard work is to provide additional Production Phase requirements and details for compliance with GCP-59, RTX Lifecycle Program Management (RLPM).

2 Persons Responsible

2.1 Scope

2.1.1 The "Production" Phase (Figure 1) intent is:

a. Manufacturing and deploying a product, software, and/or system that meets operational requirements. It includes two key efforts:

i. The production of the product and/or system potentially starting at a limited rate, allowing for further testing, risk mitigation, process refinement, and ramp up to full production.

ii. Service and support for the product and/or system.

b. Entrance criteria include validating product and processes, meeting cost targets, appropriate resource planning, incorporating lessons learned, and achieving required readiness levels. The purpose is to fulfill customer commitments and requirements.

2.1.2

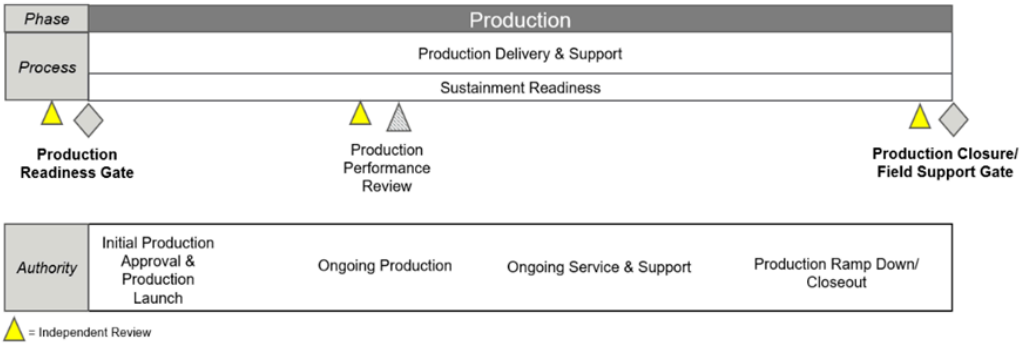

Figure 1: Production Phase Framework

2.1.3 Gates and Milestone Reviews Determination

a. No RTX-required Gates within the Production Phase

i. Production Readiness precedes entry into Production

ii. Production Closure/Field Support precedes Sustainment hand off

2.1.4 Milestone Review prior to support the Production Phase.

a. Pre-Production Readiness Review

2.1.5 Milestone Review during the Production Phase

a. Production Performance Review

2.1.6 Supplemental Reviews

a. Delta Production Readiness Review, as needed

b. Make/Buy/Where Review, as needed

2.2 Applicability

2.2.1 This standard work applies to all:

a. Organizations within RTX, including its major Business segments (each, a "**Business**"), divisions, and direct and indirect subsidiaries;

b. Majority-owned or controlled joint ventures and other majority-owned or controlled entities and operations; and

c. Directors, officers, employees, contractors, and any other persons who work for these organizations.

2.2.2 Each Business must implement the provision of this standard work within 6 months following the initial effective date. "Implement" in this context means the Business specific program lifecycle management policies are updated and released in compliance to RLPM.

2.2.3 This standard work applies throughout the RLPM Production Phase. See Exhibit A of GCP-59.

2.3 Persons Affected

2.3.1 This standard work applies to all stakeholders listed in the Responsible, Accountable, Support, and Inform (RASI) Matrix. See Exhibit A.

2.4 Governance

2.4.1 See FSW-OPSC-POL-0, Operations & Supply Chain Procedures for GCP-0 Corporate Function Requirements and Corporate Level Policy-related Functional Standard Work for governance of this FSW.

3 Responsibilities

3.1 The responsibility assignment matrix inclusive of Responsible, Accountable, Support, and Inform represents responsibilities for activities. For a successful Lifecycle Program Management, each RTX function plays a role in meeting a requirement. See Exhibit A for the RASI that represents functional assignment recommendations.

3.2 In some cases, individuals within certain functions may perform multiple roles.

4 Process Or Procedure

4.1 Requirements

4.1.1 Key Processes Associated with Production. See definitions in RTX Operations and Supply Chain Matrix.

a. For Production processes, the Operations Lead must ensure existence of:

i. Execution Readiness and Support

ii. Order Execution

iii. Shop Floor Management

b. For Quality processes, the Quality Assurance Lead must ensure existence of:

i. Product and Process Quality

ii. Quality Management System Requirements and Oversight

iii. Supplier Quality

c. For Planning processes, the Operations Lead must ensure existence of:

i. Production Control

ii. Materials Planning and Master Data

iii. Logistics

iv. Sales, Inventory, and Operations Planning (SIOP)

v. Master Production Scheduling

d. For Supply Chain processes, the Supply Chain Lead must ensure existence of:

i. Compliance

ii. Sourcing Management, including work transfers

iii. Supplier Relationship Management

iv. Source to Pay (Product/Non-Product)

4.1.2 Required Phase Entrance Criteria (Production Readiness Gate)

a. Completion of qualification testing and other applicable testing, and reports closed.

b. Meet producibility requirements including but not limited to safety, quality, labor cost, material cost, approved cost model that supports the design cost targets, and yield.

c. Completion of Design for Manufacturing and Assembly (DFMA) including tolerance stack-up analysis.

d. Completion of required AS9145 Production Part Approval Process (PPAP) elements such as those listed below in support of FCP-OPSC-QUAL-02, Product and Process Quality.

i. Design Records

ii. Design Risk Analysis (for example, DFMEA)

iii. Process Flow Diagram

iv. Process Failure Mode and Effects Analysis (PFMEA)

v. Control Plan

vi. Measurement System Analysis (MSA)

vii. Initial Process Capability Studies

viii. Packaging, Preservation, and Labelling Approvals

ix. First Article Inspection Report (FAIR)

x. Customer PPAP Requirements

xi. PPAP Approval Form, or equivalent

e. Documentation of manufacturing risks and technical debt, with funded mitigation plans in place and communication to the customer as required.

f. Manufacturing resources in place (to include but not limited to staffing, equipment, tooling, and fixtures) and documented as part of the approved Production Plan.

g. Release both the Technical Data Package (TDP) and Manufacturing Data Package (MDP).

h. Achievement of the Technology Readiness Level (TRL)/Manufacturing Readiness Level (MRL) level required by contract and/or the Business, whichever is greater (for

example, upon exiting the Development Phase, the assumption is TRL/MRL 8)

i. A Global Trade review conducted and updated the export/import plan, if required

j. Completion of the sustainment customer support strategy plan (for example, equipment, manuals, repair strategy, and training)

k. Gate Approval Process/Authority for Production Readiness Gate

i. Representation from Program Management, Engineering, Operations, Supply Chain, Quality Assurance, and Finance must attend and provide approval for entrance into Production.

ii. Business specific representatives from the following areas must attend and provide approval for entrance into Production: Legal/Contracts/Compliance and Aftermarket/Customer Support

4.1.3 Required Phase Exit Criteria (Production Closure/Field Support Gate)

a. Completion of all production contractual requirements

b. Plan for manufacturing resources (such as staff and equipment)

c. Plan for all government or customer furnished/owned property

d. Capture and communicate lessons learned

e. Documentation, communicate, and mitigate risks

f. Completion of updated sustainment strategy (to include safety requirements, aftermarket support plan, and inventory strategy)

g. Completion of improvement plan based on customer Market Feedback Analysis, a Customer Oriented Results & Excellence (CORE) method, as required

h. A Global Trade review conducted and update the export / import plan, if required

i. Completion of MRL assessments

4.1.4 Milestone Reviews

a. Review Attendance

i. Representatives from the following must attend: Program Management, Engineering, Operations, Supply Chain, Quality Assurance, Finance

ii. Additional attendees as specified by the Business must attend reviews within the Production Phase such as Contract, Legal, Aftermarket/Customer Support

b. Pre-Production Readiness Review

i. Description: Performed after CDR and at an appropriate time prior to the Production Readiness Gate. During this review, the Program Management Team (PMT) must ensure the release of production orders (internal or external) for validation represent the production design intent using the intended production processes defined, tested, and presumed capable through development. (For example, prior to ordering compliance flight test hardware.)

ii. Mandatory Criteria

1. Completion of CDR and open actions closed to plan.

2. Completion of Advanced Product Quality Planning (APQP) & PPAP Elements per FCP-OPSC-QUAL-02, RTX Product and Process Quality Policy.

3. Updated Production Plan to address the latest design configuration including but not limited to:
    a. Update of factory readiness and technology requirements
    b. Achievement of TRL/MRL as defined by the TRL/MRL maturation plan or funding of mitigation plan(s) in place
    c. Approval of capital needs and capital implementation plan meets resource needs per program schedule.
    d. Update to Supply Chain/Sourcing strategy and status to include any changes based on latest design configuration
    e. Completion of initial SIOP assessment and identification of constrained resources and operational risks (People, Digital Technology, or Capital Expenditures (CAPEX)) with mitigation plans.
4. Update of Cost Plans and Status, for example, Changes to strategy through development and learnings.
5. Update to the Sustainment Plan

c. Production Performance Review
    i. Description: An after-action review. During this review the PMT must conduct this review to validate the program is meeting the commitments made at the Production Readiness Gate.
    ii. Mandatory Criteria
        1. Incorporate Lessons learned from Initial Production
        2. Meet producibility requirements (as above in section 2)
        3. Completion of the Product and Process capability validation to include but not limited to Hardware, Test Equipment, and Manufacturing Processes
            a. Demonstrate targeted production rates or fund approved recovery plans
            b. Supply Chain readiness and performance meets Delta Production goals (to include but not limited to Staffing, Equipment, Tooling, and Fixtures)
        4. Completion of PPAP elements (as referred to in section 2d)
        5. Documentation of manufacturing risks and funding of mitigation plans, as appropriate.
        6. Manufacturing resources in place (to include but not limited to facility, staffing, equipment, tooling, and fixtures) and documented within the approved Production Plan.
        7. Release of both TDP and MDP revisions
        8. Achievement of the TRL/MRL level required by contract and/or the Business, whichever is greater (for example, upon exiting the Development Phase, the assumption is TRL/MRL 9)
        9. Update of the Sustainment Plan (for example, equipment, manuals, repair strategy, and training)

d. The PMT must conduct the Sustainment Readiness Review in conjunction with the Pre-Production Readiness Review, or prior to delivery of first product or service to customer,

to guide the post-delivery support/sustainment of product and/or services. See FSW-TGE-RLPM-03, RLPM Sustainment for details.

4.1.5 Recommended Independent Reviews (IR) (Figure 1)

a. Perform IRs prior to all formal manufacturing process reviews and Gate reviews

b. Perform IRs to assess manufacturing readiness level

c. Perform IRs prior to Production Readiness Gate (internal and supplier)

4.1.6 Supplemental Reviews

a. Delta Production Readiness Review

i. Completed upon award of follow up contracts/contract modifications/significant demand increases when applicable.

ii. Uses the similar criteria established for Production Readiness Gate.

iii. Significant changes may require a Delta Gate review.

b. Make/Buy/Where Reviews: Held prior to implementing any source changes (internal and external changes) as defined below.

i. When the sourcing supply changes, criteria of when this review must occur is as follows:

1. Originally manufactured or provided internally part is now for buy

2. Originally purchased part is now for make

3. When a change of the qualified source is desired or required for a make or purchased part

ii. When any new item that involves systems, sub-systems, components, services, and support equipment is necessary

4.1.7 Cadence Reviews within the Production Phase

a. The PMT must hold program production reviews regularly to address schedule, performance (quality & delivery), and cost. This must include a review of outstanding/open items from the Production Readiness Review.

b. The PMT must hold SIOP reviews regularly addressing schedule, material availability, inventory, staffing, and potential customer impact.

c. The PMT must hold production tier reviews addressing safety, people, quality, delivery, and cost.

d. The PMT holds a Corrective Action Board (CAB), as required, to resolve actions in a timely manner.

e. The PMT must hold Engineering review board/change control board prior to introducing any changes in production or supply chain.

f. The Supply Chain Lead must hold supplier performance reviews with underperforming and at-risk suppliers.

4.1.8 Production Plan

a. The PMT must develop a Production Plan.

i. A preliminary Production Plan during the Pursuit-to-Startup Phase with a completed draft at Submit/Commit Gate.

ii. A baseline Production Plan by Startup Gate.

iii. An updated Production Plan prior to the Production Readiness Gate.

b. Each Business must determine the contents in a Production Plan.

5 Revision History

5.1

Rev #	Date	Content Owner	Author	Details
1.0	10.1.24	M. Delcheccolo	C. Oban	Initial Release

6 Related Information

6.1 Definitions

6.1.1 See GCP-59 for definitions of terms used within this FSW.

6.2 Acronyms

6.2.1 **APQP** - Advanced Program Quality Planning

6.2.2 **CAB** - Corrective Action Board

6.2.3 **CAPEX** - Capital Expenditures

6.2.4 **CORE** - Customer Oriented Results & Excellence

6.2.5 **DFMA** - Design for Manufacturing and Assembly

6.2.6 **DFMEA** - Design Failure Mode and Effects Analysis

6.2.7 **FAIR** - First Article Inspection Report

6.2.8 **IR** - Independent Review

6.2.9 **MDP** - Manufacturing Data Package

6.2.10 **MRL** - Manufacturing Readiness Level

6.2.11 **MSA** - Measurement System Analysis

6.2.12 **PFMEA** - Process Failure Modes and Effects Analysis

6.2.13 **PMT** - Program Management Team

6.2.14 **PPAP** - Production Part Approval Process

6.2.15 **RASI** - Responsible, Accountable, Support, and Inform

6.2.16 **RLPM** - RTX Lifecycle Program Management

6.2.17 **SIOP** - Sales, Inventory & Operational Planning

6.2.18 **TDP** - Technical Data Package

6.2.19 **TRL** - Technology Readiness Level

6.3 Exhibits

6.3.1 Exhibit A - Responsibility Assignment Matrix (RASI)

6.3.2 Based on roles and responsibilities assigned within the Business, the responsible stakeholder may vary from the RASI matrix.

6.3.3

Stakeholders (Role) / Phase or Key Deliverables	Program Manager	Strategy	Business Development/Capture	Engineering	Operations	Supply Chain	Aftermarket/Customer Support	Quality Assurance	Legal/Contracts/Compliance	Finance	Other functional representatives, as required
Production Overall	A	I	I	S	**R**	S	S	S	S	S	S
Production											
Execution & Readiness Support	A	I	I	S	**R**	S	I	S	I	S	S
Order Execution	A	I	I	I	**R**	S	I	S	I	I	S
Shop Floor Management	A	I	I	S	**R**	I	I	S	I	I	I
Quality											
Product & Process Quality	A	I	I	S	S	S	I	**R**	I	I	S
Quality Management System Requirements & Oversight	A	I	I	S	S	I	I	**R**	I	I	I
Supplier Quality	A	S	S	I	I	S	I	**R**	I	I	I
Planning											
Production Control	A	I	I	I	**R**	I	I	S	I	I	I
Materials Planning & Master Data	A	I	I	I	**R**	S	I	I	I	I	I
Logistics	A	I	I	I	**R**	S	I	I	I	I	I
Sales, Inventory & Operations Planning	A	I	I	S	**R**	S	S	S	I	I	I
Master Production Schedule	A	I	I	I	**R**	S	I	I	I	I	I
Supply Chain											
Compliance	A	I	I	I	S	**R**	I	S	S	I	I
Source Strategy & Management *	A	S	S	S	S	**R**	I	S	I	I	I
Supplier Relationship Management	A	I	I	S	S	**R**	I	S	I	I	I
Source to Pay (non-Product)	A	I	I	I	I	**R**	I	I	I	I	I
Source to Pay (Product)	A	I	I	S	S	**R**	I	S	I	I	I
* Work Transfers	A	S	S	S	S	**R**	I	S	I	I	S

6.3.4 All Work Transfers follow a gated work transfer review process.

a. Legend

i. Responsible (R): Obligation to complete the deliverable or task

ii. Accountable (A): Has final decision-making authority and accountability for completion of the deliverable or task

iii. Support (S): Provides the support, assistance, and consultation for those responsible for the deliverable or task

iv. Informed (I): Informed after a decision or action

6.4 Links to reference material

6.4.1 GCP-59 RTX Lifecycle Program Management

6.4.2 FSW-OPSC-POL-0 Operations & Supply Chain Procedures for GCP-0 Corporate Function Requirements and Corporate Level Policy-related Functional Standard Work

6.4.3 FSW-TGE-RLPM-01 RLPM Pursuit-to-Startup

6.4.4 FSW-TGE-RLPM-02 RLPM Development

6.4.5 FSW-TGE-RLPM-04 RLPM Sustainment

7 Change Management

7.1 Communication, awareness and access to the FSW are key to success.

7.1.1 Communicate and train employees to perform tasks as defined in the FSW

7.1.2 Post the FSW in a location accessible by all impacted persons (examples include SharePoint Online, Teams, and ePolicy)

7.1.3 Periodically review and update the document to incorporate improvements.